

Q1 (20 July 2021 Shift 1)

Consider a mixture of gas molecule of types A, B and C having masses $m_A < m_B < m_C$. The ratio of their root mean square speeds at normal temperature and pressure is :

(1) $v_A = v_B = v_C = 0$

(2) $\frac{1}{v_A} > \frac{1}{v_B} > \frac{1}{v_C}$

(3) $v_A = v_B \neq v_C$

(4) $\frac{1}{v_A} < \frac{1}{v_B} < \frac{1}{v_C}$

Q2 (20 July 2021 Shift 2)

The correct relation between the degrees of freedom f and the ratio of specific heat γ is :

(1) $f = \frac{2}{\gamma-1}$

(2) $f = \frac{2}{\gamma+1}$

(3) $f = \frac{\gamma+1}{2}$

(4) $f = \frac{1}{\gamma+1}$

Q3 (22 July 2021 Shift 1)

What will be the average value of energy for a monoatomic gas in thermal equilibrium at temperature T ?

(1) $\frac{2}{3}k_B T$

(2) $k_B T$

(3) $\frac{3}{2}k_B T$

(4) $\frac{1}{2}k_B T$

Q4 (25 July 2021 Shift 1)

For a gas $C_P - C_V = R$ in a state P and

$C_P - C_V = 1.10R$ in a state Q, T_P and T_Q are the

temperatures in two different states P and Q respectively. Then

Questions with Answer Keys

MathonGo

(1) $T_P = T_Q$

(2) $T_P < T_Q$

(3) $T_P = 0.9 T_Q$

(4) $T_P > T_Q$

Q5 (25 July 2021 Shift 2)

Two spherical soap bubbles of radii r_1 and r_2 in vacuum combine under isothermal conditions. The resulting bubble has a radius equal to :

(1) $\frac{r_1 r_2}{r_1 + r_2}$

(2) $\sqrt{r_1 r_2}$

(3) $\sqrt{r_1^2 + r_2^2}$

(4) $\frac{r_1 + r_2}{2}$

Q6 (25 July 2021 Shift 2)

A system consists of two types of gas molecules A and B having same number density $2 \times 10^{25}/\text{m}^3$. The diameter of A and B are 10 Å and 5 Å respectively. They suffer collision at room temperature. The ratio of average distance covered by the molecule A to that of B between two successive collision is $\times 10^{-2}$

Q7 (27 July 2021 Shift 1)

The number of molecules in one litre of an ideal gas at 300 K and 2 atmospheric pressure with mean kinetic energy 2×10^{-9} J per molecules is :

(1) 0.75×10^{11}

(2) 3×10^{11}

(3) 1.5×10^{11}

(4) 6×10^{11}

Answer Key**Q1 (4)****Q2 (1)****Q3 (3)****Q4 (4)****Q5 (3)****Q6 (25)****Q7 (3)**

Hints and Solutions

MathonGo

Q1

$$V_{\text{RMS}} = \sqrt{\frac{3RT}{M}}$$

$$m_A < m_B < m_C$$

$$\Rightarrow V_A > V_B > V_C$$

$$\Rightarrow \frac{1}{V_A} < \frac{1}{V_B} < \frac{1}{V_C}$$

Q2

$$\gamma = 1 + \frac{2}{f}$$

$$f = \frac{2}{\gamma - 1}$$

Q3

As per Equi-partition law :

Each degree of freedom contributes

$$\frac{1}{2} k_B T \text{ Average Energy}$$

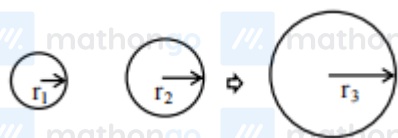
In monoatomic gas D.O.F. = 3

$$\Rightarrow \text{Average energy} = 3 \times \frac{1}{2} k_B T = \frac{3}{2} k_B T$$

Q4

 $C_P - C_V = R$ for ideal gas and gas behaves as ideal gas at high temperature so $T_P > T_Q$

Q5



no. of moles is conserved

$$n_1 + n_2 = n_3$$

$$P_1 V_1 + P_2 V_2 = P_3 V$$

$$\frac{4S}{r_1} \left(\frac{4}{3} \pi r_1^3 \right) + \frac{4S}{r_2} \left(\frac{4}{3} \pi r_2^3 \right) = \frac{4S}{r_3} \left(\frac{4}{3} \pi r_3^3 \right)$$

Hints and Solutions

MathonGo

$$r_1^2 + r_2^2 = r_3^2$$

$$r_3 = \sqrt{r_1^2 + r_2^2}$$

Q6

 $\therefore$ mean free path

$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{d_2^2 n_2}{d_1^2 n_1}$$

$$= \left(\frac{5}{10} \right)^2 = 0.25 = 25 \times 10^{-2}$$

Q7

$$KE = \frac{3}{2} kT$$

$$PV = \frac{N}{N_A} RT$$

$$N = \frac{PV}{kT}$$

$$= N = 1.5 \times 10^{11}$$